SMART WORKPLACE SAFETY INTELLIGENCE PLATFORM

Comprehensive Project Specification, Architectural Principles & Academic Defense Framework

PHASE 1 (40% IMPLEMENTATION) | AI VIDEO MONITORING, PPE DETECTION & OBSTACLE ANALYSIS

1. Executive Summary & Problem Statement

Industrial construction environments, manufacturing production lines, and logistics warehouses represent high-risk operational zones. According to occupational health and safety authorities (OSHA and ILO), workplace incidents caused by non-compliance with Personal Protective Equipment (PPE) and unexpected physical obstructions along designated walkways account for thousands of preventable injuries and fatalities each year. Traditional safety enforcement relies on periodic manual inspections or post-incident forensic CCTV reviews, which fail to prevent accidents in real time.

The **Smart Workplace Safety Intelligence Platform** transitions workplace safety from a *passive post-incident audit model* into an *autonomous, proactive, real-time safety intelligence system*. Powered by state-of-the-art Computer Vision (YOLOv8) and Deep Learning, the platform continuously analyzes video streams, verifies that workers are equipped with required PPE (helmets and high-visibility safety vests), and detects hazardous physical obstacles inside safety transit zones before dangerous accidents happen.

STRICT IMPLEMENTATION SCOPE: This project adheres to rigorous phased engineering. The current operational platform covers exactly the **FIRST 40% (Phase 1)** of the master product vision: Real-Time Video Ingestion, Worker Detection, Helmet & Vest Compliance Analysis, Safety Zone Geometry, Temporal Obstacle Persistence Tracking, and Live WebSocket Dashboard Streaming. The remaining ~60% (Fall Detection, IoT Sensor Fusion, Advanced Predictive Analytics, Privacy Blur) is reserved for Phases 2–5.

2. Master Scope Matrix (Phase 1 vs Remaining 60%)

Capability Module	Description & Functional Scope	Phase	Current Status
A–C. Video & Worker Detection	Ingestion from webcam/file, frame decoding, worker identification via YOLOv8n	Phase 1	**COMPLETED**

D–H. PPE Detection & Compliance	Spatial IoU matching of helmets & safety vests to workers with 4 compliance states	Phase 1	**COMPLETED**
I. Initial Obstacle Detection	Geometric intersection of objects in defined safety zones with temporal persistence	Phase 1	**COMPLETED**
M. Safety-Zone Monitoring	Interactive user definition of walkway zones; persistence frame thresholding	Phase 1	**COMPLETED**
R–W. Event Engine & Logging	Automatic event trigger, cooldown deduplication, SQLite storage & snapshot capture	Phase 1	**COMPLETED**
Real-Time Dashboard & WebSockets	Live browser canvas rendering with bounding box overlays, FPS, latency profiling	Phase 1	**COMPLETED**
J–K. Fall & Posture Detection	MediaPipe / Pose estimation for worker falls, prostration & dangerous posture	Phase 2	**PLANNED (60%)**
L. Restricted-Area Intrusion	Worker tracking into hazardous machinery zones with virtual geofencing	Phase 2	**PLANNED (60%)**
N–Q. IoT Sensor Fusion	Hardware telemetry for gas leaks (MQ-2), smoke detection & abnormal temperature	Phase 3	**PLANNED (60%)**
X–Y. Advanced Analytics	Compliance heatmaps, frequency trend analysis, incident prediction & automated PDF reports	Phase 4	**PLANNED (60%)**
Z–AB. Privacy & Enterprise Scale	Automated face anonymization/blurring, multi-camera RTSP matrix, RBAC authentication	Phase 5	**PLANNED (60%)**

3. Core Functional Modules Deep-Dive

3.1 Video Ingestion & Frame Normalization Subsystem

The video ingestion pipeline (`app/video/processor.py`) utilizes a dedicated OpenCV capture worker. To prevent UI thread and asynchronous event-loop starvation in FastAPI, frame decoding runs on a separate executor thread. The system accepts both recorded high-definition video files (.mp4, .avi, .mkv) and real-time live webcam feeds (USB index 0). Incoming video frames are dynamically resized to 640×640 pixels with aspect-ratio preservation and configurable frame skipping to maintain steady real-time throughput even on resource-constrained hardware.

3.2 PPE Compliance Analysis Engine (Spatial Association)

A critical limitation of superficial safety systems is treating equipment detection independently from person detection (e.g., detecting a helmet resting on a desk and reporting compliance). Our PPE Analyzer (`app/detection/ppe_analyzer.py`) implements a **hierarchical spatial association algorithm**:

- **Step 1 (Worker Localization):** Filter all bounding boxes classified as 'person' with confidence ≥ 0.50.
- **Step 2 (Safety Gear Isolation):** Detect all instances of 'helmet' and 'safety-vest' exceeding confidence thresholds.
- **Step 3 (Spatial IoU Matching):** For each worker, calculate Intersection-over-Union (IoU) and proximity bounds against gear detections. Helmets must spatially correlate with the head/upper quadrant of the person box, while safety vests must cover the torso quadrant.
- **Step 4 (Granular State Decision):** The engine outputs four distinct compliance states: **1) COMPLIANT** (both helmet and vest verified), **2) HELMET_MISSING** (vest verified, helmet missing), **3) VEST_MISSING** (helmet verified, vest missing), and **4) VIOLATION** (neither equipment item present).

3.3 Obstacle Detection & Safety Zone Engine (The Phase 1 Core)

Workplace obstacle detection is the signature capability developed in Phase 1. Rather than classifying every stationary object in the workspace as an obstacle, the system applies rigorous spatial-geometric reasoning (`app/detection/obstacle_analyzer.py` & `app/detection/tracker.py`):

- **Geometric Safety Zones:** Administrators interactively define 2D polygon safety zones over the camera feed (such as 'Primary Walkway', 'Forklift Transit Lane'). Coordinates are stored as normalized vectors [0.0 – 1.0] to remain resolution-independent.
- **Exclusion Filtering:** Human workers and authorized safety equipment are excluded from obstacle candidate sets.
- **Polygon Overlap Calculation:** The candidate object's bounding box is converted into a geometric polygon using **Shapely**. The intersection area between the object box and the active zone polygon is computed: Overlap Ratio = Area(Box ∩ Zone) / Area(Box). If Overlap Ratio ≥ 0.30 (30%), a potential incursion is registered.
- **Temporal Persistence Verification:** Single-frame detection glitches (such as transient shadow or sensor flicker) do *not* trigger alarms. The spatial tracking module clusters detections across frames. An obstacle must persist for at least **N consecutive frames** (default: 10 frames ≈ 0.35–1.0s) before an incident is officially generated.
- **Event Cooldown & Deduplication:** A 30-second cooldown per object/zone prevents database saturation while maintaining an active on-screen visual warning.

4. Academic Defense Framework (Viva / Review Examination Table)

During internal evaluations, project reviews, and final viva examinations, each implemented feature can be defended across the standard academic examination rubric defined below:

Rubric	PPE Compliance Detection	Workplace Obstacle Detection	Real-Time Monitoring Canvas
WHAT	Identifies if workers wear helmets & vests in live video.	Detects hazardous physical clutter inside walkways.	Renders live video with bounding boxes & safety telemetry.
WHY	Non-compliance causes preventable head & torso trauma.	Walkway clutter is the #1 cause of industrial trip-and-falls.	Supervisors need instant visual situational awareness.
HOW	YOLOv8 bounding box detection + IoU spatial pairing.	Shapely polygon intersection + temporal frame tracking.	FastAPI WebSockets + HTML5 Canvas rendering.
INPUT	640×640 RGB video frames from camera or file.	Normalized safety zone polygons + object bboxes.	Base64 JPEG stream + telemetry JSON payload.
OUTPUT	4 compliance states (Compliant, Helmet/Vest Missing, Violation).	Hazard alert box, zone ID, frame persistence count.	Real 30+ FPS UI canvas, live latency (ms), true FPS.
LIMITATION	Extreme distance or severe occlusions lower gear recall.	Rectangular zones in Phase 1 UI; complex curves pending.	Single camera active simultaneously in Phase 1.
FUTURE	Gloves, goggles, harnesses via Phase 2 pose/vision model.	3D depth camera integration for volumetric obstruction.	Multi-camera RTSP matrix with distributed edge nodes.

5. Remaining 60% Roadmap (Phases 2 to 5)

The remaining 60% of the platform architecture has been fully designed and stubbed with clean interfaces, ready for immediate sequential execution upon explicit user authorization:

Phase & Allocation	Planned Capabilities	Technical Implementation Approach
Phase 2 (60–70%) Advanced Vision & Behavior	• Worker Fall Detection • Dangerous Posture Recognition • Restricted Area Geofencing • Advanced PPE (Gloves, Goggles, Masks)	YOLOv8-Pose / MediaPipe 33-keypoint skeletal tracking; temporal body axis orientation; polygon ray-casting containment tests.
Phase 3 (70–80%) IoT Environmental Fusion	• Hazardous Gas Leak Alerts • Industrial Smoke Detection • Extreme Temperature & Humidity • Multi-modal Sensor Fusion	ESP32 / Raspberry Pi microcontroller nodes transmitting via MQTT / Webhooks; MQ-2 gas sensors, optical smoke detectors, DHT22 sensors.
Phase 4 (80–90%) Analytics & Reporting	• Automated Safety Audits • Incident Density Heatmaps • PPE Compliance Trends • Predictive Risk Forecasting	Pandas/NumPy time-series aggregation; kernel density estimation (KDE) heatmaps; automated PDF executive compliance reports.
Phase 5 (90–100%) Enterprise Privacy & Scale	• Facial Anonymization & Privacy • Multi-Camera RTSP Management • Role-Based Access Control (RBAC) • Edge TensorRT Acceleration	Automated facial Gaussian blurring/masking; OpenCV GStreamer RTSP matrix; JWT authentication & OAuth2; NVIDIA TensorRT FP16 engine export.